DO NOW — RECALL

Lesson 109 · CPU Scheduling: FCFS, SJF and SRT

Name: _____ Class: _____ Date: _____

1. Name the routine executed when an interrupt is accepted.

2. An OR gate has inputs 0 and 1. What is the output?

3. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

4. What is the time complexity of sequential search in the worst case?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Interrupt service routine (ISR).

2. Model answer

1.

3. Model answer

X is removed; Y remains at the front.

4. Model answer

O(n).

By the end, you can…

• First come, first served: non-preemptive queue in arrival order.

• Shortest job first: choose least burst time from ready jobs; non-preemptive.

• Shortest remaining time: preempt if a new job has a shorter remaining burst.

Vocabulary — know these terms

FCFS — Jobs run in first-arrival order.

SJF — Selects the shortest estimated ready job.

SRT — Preemptive shortest remaining time.

waiting time — Time spent ready but not executing.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Order B 0–2, C 2–5, A 5–11. Waits B 0, C 2, A 5; mean 7/3 ≈ 2.33 units. SJF assumes burst-length estimates are available.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

FCFS

First come, first served: non-preemptive queue in arrival order.

SJF

Shortest job first: choose least burst time from ready jobs; non-preemptive.

SRT

Shortest remaining time: preempt if a new job has a shorter remaining burst.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

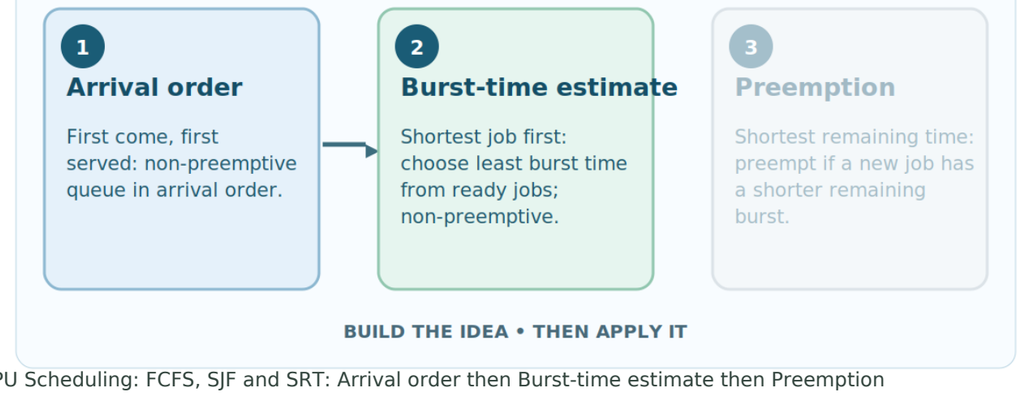
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CPU Scheduling: FCFS, SJF and SRT: Arrival order then Burst-time estimate then Preemption

New Learning



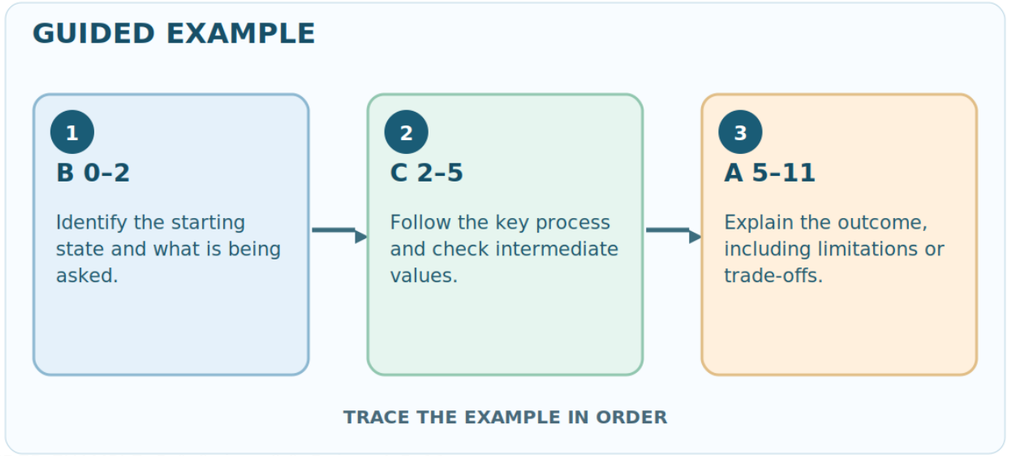
CPU Scheduling: FCFS, SJF and SRT: Arrival order then Burst-time estimate then Preemption

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

SJF with all ready at 0: A burst 6, B burst 2, C burst 3. Trace and calculate mean waiting time.

Identify the inputs, relevant facts and exact output or judgement required.



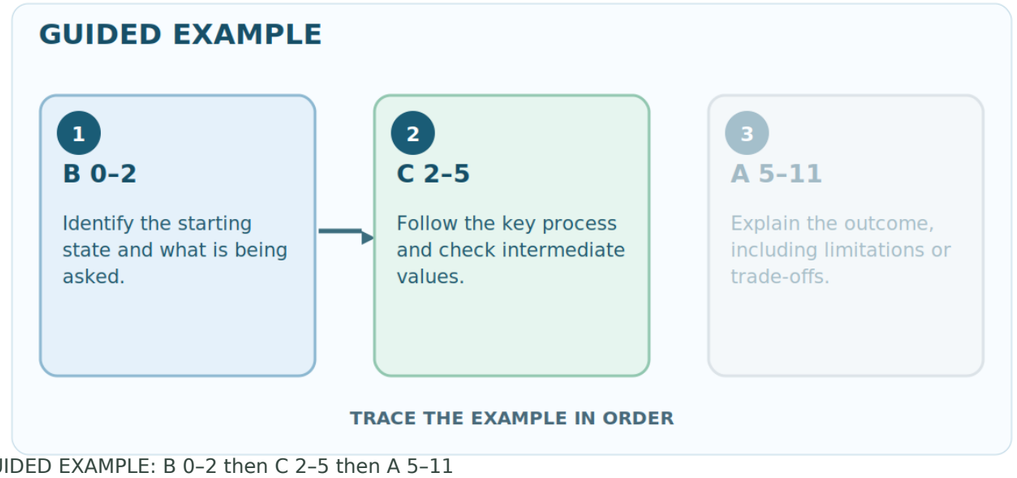
GUIDED EXAMPLE: B 0–2 then C 2–5 then A 5–11

We Do Check · original lesson diagrams

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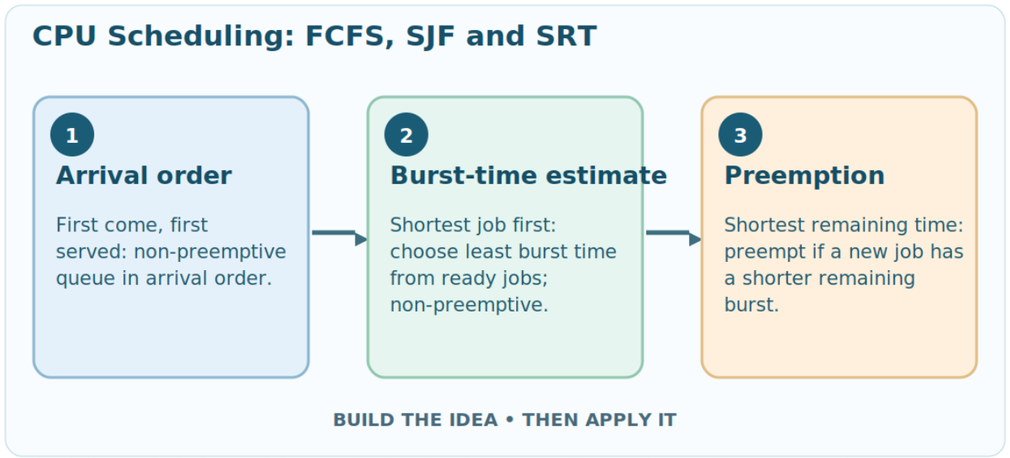


GUIDED EXAMPLE: B 0–2 then C 2–5 then A 5–11

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CPU Scheduling: FCFS, SJF and SRT: Arrival order then Burst-time estimate then Preemption

I DO — WORKED EXAMPLE

Follow the worked method

FCFS: P1 arrives at 0 (burst 4), P2 at 1 (burst 2), P3 at 2 (burst 1). Find waiting times.

Model answer

Timeline P1 0–4, P2 4–6, P3 6–7. Waiting = start − arrival: P1 0, P2 3, P3 4. Average = 7/3 ≈ 2.33 units.

Check: link each claim to the scenario and use precise OCR terminology.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

SRT never interrupts an executing process.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

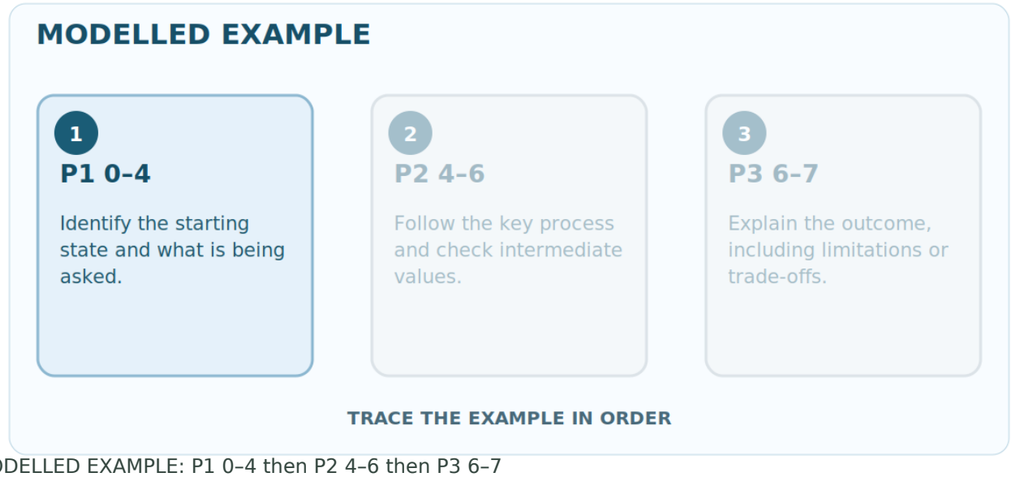
SRT: A arrives 0/burst 5, B arrives 1/burst 2. Show the timeline and explain preemption. [8]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

FCFS: P1 arrives at 0 (burst 4), P2 at 1 (burst 2), P3 at 2 (burst 1). Find waiting times.

Timeline P1 0–4, P2 4–6, P3 6–7. Waiting = start − arrival: P1 0, P2 3, P3 4. Average = 7/3 ≈ 2.33 units.



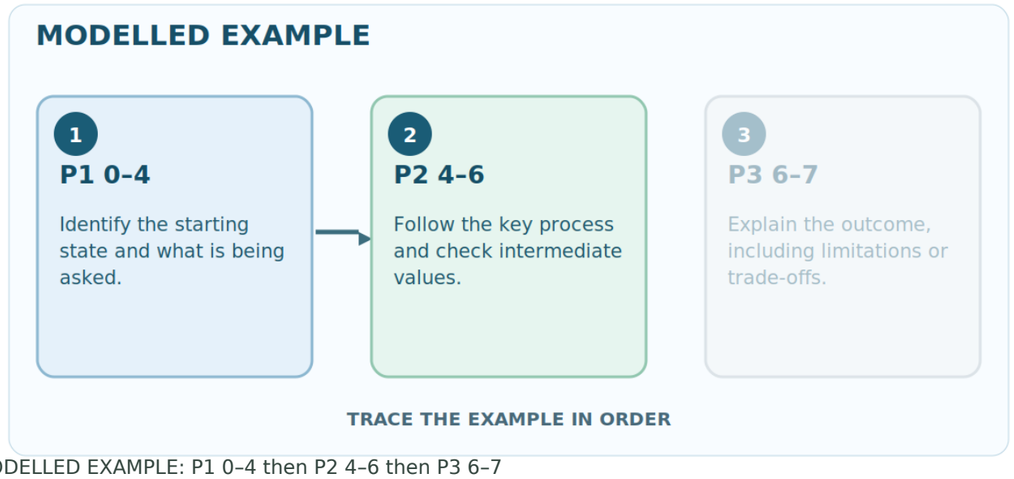
MODELLED EXAMPLE: P1 0–4 then P2 4–6 then P3 6–7

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

FCFS: P1 arrives at 0 (burst 4), P2 at 1 (burst 2), P3 at 2 (burst 1). Find waiting times.

Timeline P1 0–4, P2 4–6, P3 6–7. Waiting = start − arrival: P1 0, P2 3, P3 4. Average = 7/3 ≈ 2.33 units.



MODELLED EXAMPLE: P1 0–4 then P2 4–6 then P3 6–7

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Distinguish non-preemptive and preemptive scheduling and give an example of each. [6]

Question 2

FCFS: A arrives 0/burst 3, B arrives 1/burst 2, C arrives 2/burst 4. Calculate each wait. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

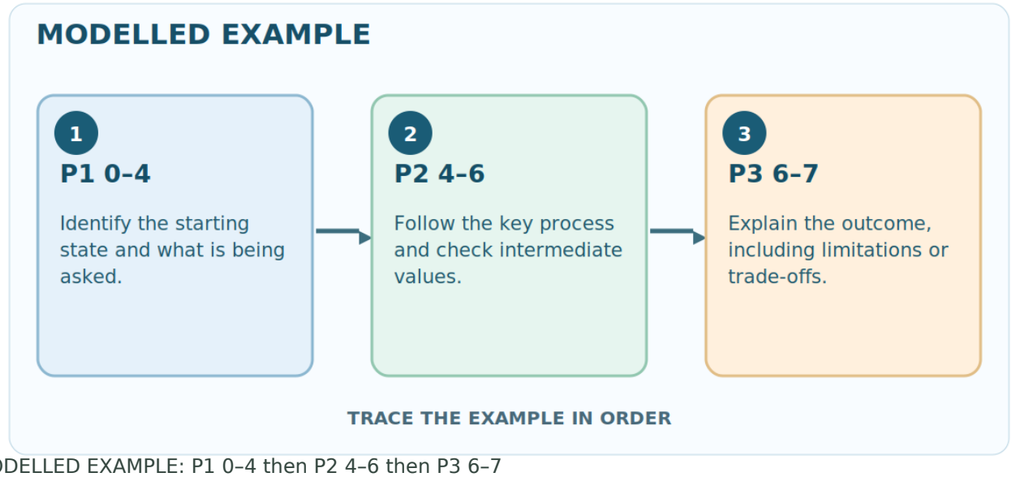
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

FCFS: P1 arrives at 0 (burst 4), P2 at 1 (burst 2), P3 at 2 (burst 1). Find waiting times.

Timeline P1 0–4, P2 4–6, P3 6–7. Waiting = start − arrival: P1 0, P2 3, P3 4. Average = 7/3 ≈ 2.33 units.



MODELLED EXAMPLE: P1 0–4 then P2 4–6 then P3 6–7

WE DO — GUIDED PRACTICE

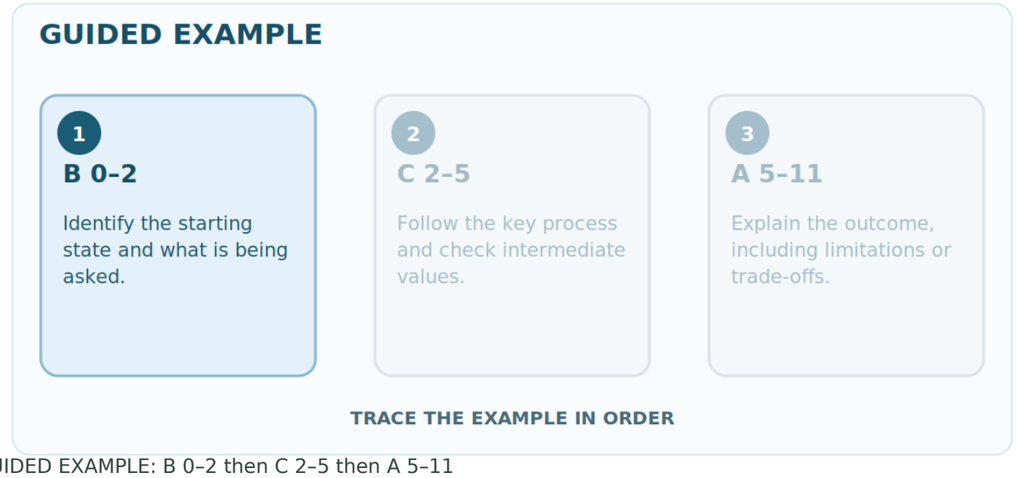
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

SJF with all ready at 0: A burst 6, B burst 2, C burst 3. Trace and calculate mean waiting time.

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: B 0–2 then C 2–5 then A 5–11